

Session 5 · Deck 5.A4

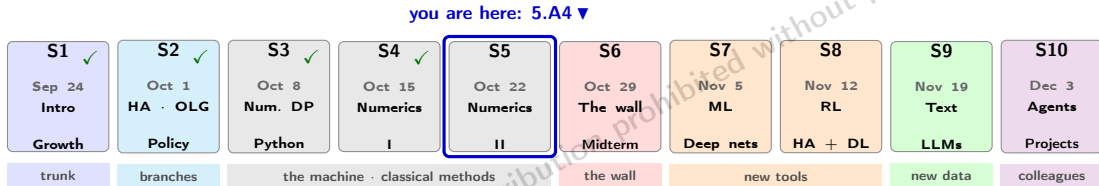
Linear Rational-Expectations Systems

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Thursday, October 22, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 5.A1 what perturbation computes · 5.A2 the model and the parameter · 5.A3 first order · **5.A4** linear RE systems · 5.A5 higher order, risk, pruning · 5.A6 the worked example.

Linearize, or Log-Linearize?

Both are first-order Taylor expansions; they differ in the variable.

Log-linearization. Substitute $x_t = x e^{\hat{x}_t}$ where

$$\hat{x}_t = \ln \frac{x_t}{x},$$

and expand in $\hat{x}_t$. Then every coefficient is an **elasticity** and every deviation is a **percentage**, which is how macro results are reported anyway.

Linearization. Expand in $x_t - x$ directly. Coefficients carry units.

They are the same approximation under a change of variables, so you may do either and convert afterwards — but the accuracy is *not* identical, because the two are exact for different functions.

Which is better here?

5.A2 settled it for this model: the Brock–Mirman policy is **exactly linear in logs** and curved in levels. Log-linearizing recovers it exactly; linearizing does not.

That is a property of this model, not a law. Logs are the right default in macro because most variables are positive and grow.

Linearizing the Equilibrium Conditions

Take the quarterly model of 5.A3 and expand each condition around $(k^*, c^*, 0)$, writing hats for log deviations.

Resource constraint.

$$k^* \hat{k}_{t+1} = (\alpha y^* + (1 - \delta)k^*) \hat{k}_t + y^* z_t - c^* \hat{c}_t$$

Euler equation.

$$\mathbb{E}_t \hat{c}_{t+1} = \hat{c}_t + \kappa \mathbb{E}_t z_{t+1} + \kappa(\alpha - 1) \hat{k}_{t+1}$$

with $\kappa = \beta \alpha (k^*)^{\alpha-1}$ — the share of the gross return that actually moves. Measured: $\kappa = 0.03475$.

Shock. $\mathbb{E}_t z_{t+1} = \rho z_t$.

Structure follows FV&G, *Solution Methods V* (Appendix), “Linearization II–III”; instantiated on this course’s model rather than theirs, which carries endogenous labour. Ledger pert_first_order.

Why κ is small

At $\delta = 0.025$, most of next period’s gross return is undepreciated capital, which does not respond to z or k . Only 3.5% of it does. That single number is why capital is so persistent here, and why the model propagates shocks for years.

Rewriting It as a System

After collecting terms, any linearized model has the shape

$$\begin{aligned} A \hat{s}_{t+1} + B \hat{s}_t + C \hat{y}_t + D z_t &= 0 \\ \mathbb{E}_t (F \hat{s}_{t+1} + G \hat{s}_t + H \hat{s}_{t-1} \\ + J \hat{y}_{t+1} + K \hat{y}_t + L z_{t+1} + M z_t) &= 0 \\ \mathbb{E}_t z_{t+1} &= N z_t \end{aligned}$$

with m states $\hat{s}_t$, n controls $\hat{y}_t$, and k exogenous processes z_t .

The first block is the conditions that hold **within** a period; the second is the ones involving **expectations**. That split is the whole difficulty: an ordinary difference equation would already be solved.

FV&G, *Solution Methods V* (Appendix), "General structure of linearized system".

You may eliminate controls by substitution before forming the system, as FV&G do with c_t . It is not necessary, and it changes only the arithmetic — not the policy functions you end up with.

N must have only stable eigenvalues: the exogenous process has to be stationary, or nothing below works.

Undetermined Coefficients: Guess, Substitute, Match

Guess that the solution is linear:

$$\hat{s}_t = P \hat{s}_{t-1} + Q z_t$$

$$\hat{y}_t = R \hat{s}_{t-1} + U z_t$$

Substitute into the system, use linearity of expectations and $\mathbb{E}_t z_{t+1} = N z_t$, and collect.

Because the result must hold for **every** value of $\hat{s}_{t-1}$ and z_t , each coefficient must vanish separately. In the scalar case that gives, on $\hat{s}_{t-1}$,

$$AP_1 + B + CR_1 = 0, \quad GP_1 + H + JR_1P_1 + KR_1 = 0,$$

and two more on z_t .

FV&G, *Solution Methods* V (Appendix), “Undetermined coefficients” and “Solving the system I”.

The same trick as 5.A3

“It holds identically, so match coefficients.” There we matched derivatives of F ; here we match powers of the state. Same idea, different bookkeeping.

Four equations, four unknowns (P_1, Q, R_1, U) — and, crucially, the first pair does *not* involve Q or U . The state block solves on its own.

The Quadratic, and Its Two Roots

Eliminating $R_1 = -C^{-1}(AP_1 + B)$ from the pair leaves a single **quadratic** equation in P_1 :

$$P_1^2 + \left(\frac{B}{A} + \frac{K}{J} - \frac{GC}{JA} \right) P_1 + \frac{KB - HC}{JA} = 0.$$

It has two roots,

$$P_1 = \frac{1}{2} \left[-(\cdot) \pm \sqrt{(\cdot)^2 - 4 \frac{KB - HC}{JA}} \right],$$

one stable and one unstable. Keep the stable one, recover R_1 , and Q and U then follow from two linear equations.

This is where the “quadratic system” of 5.A3 turns into an actual choice — and why counting equations was never enough.

FV&G, *Solution Methods V* (Appendix), “Solving the system II–III”; and *Solution Methods V*, “Solving the system II” (“Why quadratic? Stable and unstable manifold”).

Why quadratic at all?

Because the model has a **stable and an unstable manifold**. The two roots are those two manifolds; discarding the explosive one is imposing the transversality condition.

1.B1's saddle path, in algebra. There we drew it; here we solve for it.

In Matrices: One Quadratic Equation

With more than one state the same derivation gives a **matrix** quadratic. P solves

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P - KC^{-1}B + H = 0,$$

and the equilibrium is stable if and only if

$$\max |\text{eig}(P)| < 1.$$

The rest is linear algebra:

$$R = -C^{-1}(AP + B), \quad U = -C^{-1}(AQ + D),$$

with Q from a Sylvester equation in $\text{vec}(Q)$.

FV&G, *Solution Methods V* (Appendix), "Policy functions II". Uhlig's toolkit covers the general $l > n$ case.

Everything about the model is now in **one matrix equation** and **one eigenvalue condition**. The economics has been fully absorbed into $A, \dots, N$.

C must be invertible (rank n). When it is not — more static conditions than controls — you need the general case, which is the QZ route on the next slide.

How to Solve a Matrix Quadratic

To solve $\Psi P^2 - \Gamma P - \Theta = 0$ for the $m \times m$ matrix P :

1. Build the $2m \times 2m$ pair

$$\Xi = \begin{bmatrix} \Gamma & \Theta \\ I_m & 0_m \end{bmatrix}, \quad \Delta = \begin{bmatrix} \Psi & 0_m \\ 0_m & I_m \end{bmatrix}.$$

2. Take the **generalized** eigenvalues ω_i and eigenvectors s_i of Ξ with respect to Δ . Each has the form $s'_i = [\omega_i x'_i, x'_i]$.

3. If m of them give linearly independent x_i , then

$$P = \Omega \Lambda \Omega^{-1}, \quad \Omega = [x_1, \dots, x_m], \quad \Lambda = \text{diag}(\omega_i),$$

solves it, and is stable iff $\max |\omega_i| < 1$.

Note the symbol

The generalized eigenvalues are written ω here. The literature writes λ ; this course reserves λ for the invariant distribution, and χ is already the perturbation parameter.

Quadratic in P , but *linear* in the pencil (Ξ, Δ) — which is exactly the trick that turns a hard nonlinear problem into a library call.

FV&G, *Solution Methods V* (Appendix), "How to solve quadratic equations". In code: `scipy.linalg.ordqz` — the generalized Schur (QZ) decomposition, which is Klein's (2000) route.

Blanchard–Kahn: The Condition

Write the linearized model as

$$A x_{t+1|t} = B x_t + C z_t, \quad z_{t+1} = \rho z_t + \varepsilon_{t+1},$$

where x_t stacks states and controls. Split the variables:

- **Predetermined** — capital. Cannot jump; its value today was decided yesterday.
- **Non-predetermined** (“jump”) — consumption. Chooses itself now, and can move discontinuously.

Blanchard–Kahn condition. There is a unique stable solution if and only if

$$\#\{|\omega_i| > 1\} = \#\{\text{jump variables}\}.$$

Blanchard & Kahn (1980), *Econometrica* 48(5). Frame after Quant_Macro Lec. 9, “The Blanchard–Kahn conditions”.

The intuition

Each unstable root is a direction along which the economy would explode. Each jump variable is one instrument for stopping it — choose today's value to sit on the stable manifold. Too few instruments and paths explode; too many and the extra ones are free.

The Three Cases

Count	Name	What it means
unstable roots = jumps	Uniqueness	Exactly one bounded path from any initial state. The normal case, and what you hope for.
unstable roots < jumps	Indeterminacy	A continuum of bounded paths. Expectations can be self-fulfilling; sunspots matter.
unstable roots > jumps	Non-existence	No bounded path at all. The model, as written, has no equilibrium of this kind.

In the growth model

One predetermined variable (k), one jump (c). So we need **exactly one** root outside the unit circle — which is the saddle-path property, restated as a count.

Indeterminacy is not a bug to be tuned away. 2.B2 met it in an OLG economy (Feng & Hoelle 2017), where it is the substantive result.

Measured: Our Two Growth Models Pass

Generalized eigenvalue moduli, computed by QZ:

Model	$ \omega $	Verdict
quarterly, $\delta = 0.025$	0.950, 0.962, 1.050	unique
Brock–Mirman, $\delta = 1$	0.360, 0.950, 2.924	unique

One root outside the unit circle in each, one jump variable in each. The condition holds.

The unstable root of the $\delta = 1$ model is 2.923977, and $1/(\alpha\beta) = 2.923977$ — the closed form again, arriving through an eigenvalue solver that knows nothing about it.

Ledger blanchard_kahn.growth_models and pert_first_order.

Note the two stable roots: $\rho = 0.95$, which is just the shock, and 0.962, which is the endogenous persistence of capital. The model's memory is *longer* than the shock's.

The unstable root here is 1.0499, **not** $1/\beta = 1.0101$. The two coincide only at $\delta = 1$.

Measured: A Model Where the Count Actually Bites

The growth model can only ever show case 1. To see the other two, take the three-equation New Keynesian model,

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t$$

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\varsigma} (i_t - \mathbb{E}_t \pi_{t+1})$$

$$i_t = \phi_\pi \pi_t$$

Two jump variables (π, x) , one exogenous state. So uniqueness needs **two** unstable roots.

Sweeping the policy rule:

ϕ_π	$ \omega $	unstable
0.0	0.704, 0.800, 1.435	1
0.8	0.800, 0.895, 1.244	1
1.0	0.800, 1.000, 1.139	1
1.5	0.800, 1.097, 1.097	2
2.5	0.800, 1.154, 1.154	2

The switch happens at $\phi_\pi = 1$: the **Taylor principle**, recovered from an eigenvalue count. Below it the equilibrium is indeterminate; at exactly 1 a root sits on the unit circle.

Ledger blanchard_kahn.new_keynesian: $\kappa = 0.1275$, $\varsigma = 1$, $\rho_u = 0.8$. For $\phi_\pi < 1$ the QZ routine cannot even form P — the block it must invert is singular, which is what indeterminacy looks like from inside the algorithm.

In Practice: Who Does This For You

Four procedures solve the quadratic system, and they are equivalent: **Blanchard & Kahn (1980)**, **Uhlig (1999)**, **Sims (2002)**, **Klein (2000)**. They differ in what they assume about the matrices, not in the answer.

In practice you use software: **Dynare** for first, second and third order; Dynare++ or Mathematica beyond that. You supply the equilibrium conditions; it does the steady state, the derivatives and the root selection.

The real burden is the **derivatives**. Dynare computes them analytically by symbolic differentiation, and for a reason.

Why not numerical derivatives?

We used them, and watched them fail. First order: the finite-difference solve matched QZ to 4.4×10^{-11} . Second order, with a step ten times smaller, the residual of the same system went from 3.6×10^{-9} to 8×10^{-3} . Second derivatives divide by h^2 , so rounding error is amplified by 10^8 — 4.A4's U-curve, in a place where it silently corrupts a published coefficient.

FV&G, *Solution Methods V*, "Solving the system II" and "Computer". The failure above is from this deck's own build log, not a source. Automatic differentiation (4.A4) is the modern answer.

Takeaways

- **Log-linearize** by default: coefficients are elasticities, deviations are percentages — and for this model it is exact.
- Guess linear rules, substitute, match coefficients: the state block collapses to a **matrix quadratic** in P , stable iff $\max |\text{eig}(P)| < 1$. It is quadratic because the model has a **stable and an unstable manifold** — discarding the explosive root is imposing transversality.
- Solve it as a **generalized eigenvalue problem**: the QZ decomposition, one library call, Klein's (2000) route.
- **Blanchard–Kahn**: unique iff ($\#$ roots outside the unit circle) = ($\#$ jump variables). Fewer gives indeterminacy, more gives non-existence.
- Measured: both growth models have exactly one unstable root (1.050 quarterly, $2.924 = 1/(\alpha\beta)$ at $\delta = 1$); in the New Keynesian model the count flips at $\phi_\pi = 1$ — **the Taylor principle as an eigenvalue count**.
- Use Dynare, and let it differentiate **analytically**: our own numerical second derivatives cost six orders of magnitude of residual.

Bridge: What the Linear Solution Cannot See

We now have a complete, unique, checkable first-order solution. It also has, by construction, **no risk in it at all**: 5.A3 showed $c_\chi = k_\chi = 0$, so the linear rules are the same whether the shock has standard deviation 0.007 or 0.

For dynamics that is often fine. For anything about uncertainty — precautionary saving, risk premia, the welfare cost of business cycles — it returns exactly zero, which is not an estimate.

Next (5.A5)

Second order: the twelve-equation system, the risk correction, and why every odd derivative in χ vanishes. Then why second-order simulations explode, and the pruning that stops them.

Also: how far from the steady state any of this is valid, which turns out to be a question about *complex* analysis.